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AWS Services Lesson

Skills you will learn

Cloud Computing: Cloud Fundamentals

> ***Why learn AWS?*** *It holds ~33% of global cloud market share. Whether you're DevOps, data, or full-stack, you'll deploy or integrate with AWS at some point. This lesson sets the conceptual ground before diving into hands-on labs.*

0 ▸ Learning Outcomes

By the end you can ✅:

1. Navigate the AWS Console, CLI, and SDK docs.
2. Sketch the AWS *global infrastructure* hierarchy from memory.
3. Choose an AWS region based on latency, cost, and compliance.
4. Explain each pillar of the **Well-Architected Framework**.
5. Map migration challenges to **AWS Cloud Adoption Framework (CAF)** perspectives.
6. Distinguish security roles in the **Shared Responsibility Model**.

1 ▸ Access Paths to AWS Services

Interface	Best for	Example
AWS Management Console	One-off tasks, beginners	Launch first EC2
AWS CLI (`aws s3 cp …`)	Scripting, CI pipelines	Sync S3 bucket
SDKs (Python boto3, JS AWS-SDK)	Programmatic control in apps	Put item into DynamoDB
Infrastructure-as-Code (CloudFormation, Terraform)	Version-controlled infra	Deploy VPC + RDS

> 🛠️ ***Try it:*** *`aws configure list` to view CLI creds after you set them.*

2 ▸ AWS Global Infrastructure Visual

```
World
└─ Region (us-east-1)
   ├─ AZ a  ─┬─ Data Centre
   │         └─ Data Centre
   ├─ AZ b  ─┬─ Data Centre
   └─ Edge Locations (CloudFront POPs)
```

- **Regions** = geographic clusters (availability, compliance).
- **AZs** = isolated campuses with <2 ms latency between them.

3 ▸ Choosing the Right Region

Factor	Check	Tool
Latency	Ping tests to endpoints	CloudPing.info
Price	Compare on-demand rates	AWS Pricing Calc
Compliance	Data residency laws (GDPR, HIPAA)	AWS Artifact reports
Service availability	Not all services in all regions	https://aws.amazon.com/about-aws/global-infrastructure/regional-prc
Sustainability	Carbon reduction targets	*Regions with hydropower*

> 🎯 ***Rule:*** *Put compute near users, but keep data where regulators require.*

4 ▸ AWS Well-Architected Framework — 6 Pillars in Plain English

Pillar	One-liner	Starter Questions
Operational Excellence	Can we deploy/recover quickly?	CI/CD? Runbooks?
Security	Is data safe & least-privileged?	IAM roles? KMS encryption?
Reliability	Will it stay up under failures?	Multi-AZ? Backup restore?
Performance Efficiency	Do we use right resources?	Autoscaling? Graviton?
Cost Optimization	Are we paying only for what we need?	Rightsizing? Spot?
Sustainability	Are we minimising carbon & waste?	Turn off idle dev boxes?

> 🧦 ***Exercise:*** *Pick a hobby app—rate it 1-5 against each pillar.*

5 ▸ AWS Cloud Adoption Framework (CAF)

CAF breaks transformation into **6 perspectives**:

1. **Business** – value, KPIs, ROI.
2. **People** – skills, roles, culture.
3. **Governance** – compliance, risk, budget.
4. **Platform** – architecture, tooling.
5. **Security** – identity, detection, response.
6. **Operations** – monitoring, incident management.

> 🧩 ***Mapping drill:*** *Where does "Automated cost-alerts" fit? (Governance).*

6 ▸ Shared Responsibility Model Cheat Card

Layer	AWS **owns**	Customer **owns**
Physical	Data centres, hardware	—
Network	Core networking, edge	VPC design, NACLs, SGs
Host OS	Managed services (Lambda, S3)	Your EC2 AMIs, patching
Application	—	Code, MFA, IAM roles
Data	—	Encryption keys, backups

Remember the rhyme: **"AWS secures OF the cloud, you secure IN the cloud."**

AWS Service Families

Category	Key Services	One-Sentence Purpose
Compute	**EC2** (Elastic Compute Cloud)	Resizable virtual machines for any OS or workload.
	Lambda	Run code in response to events—no servers to manage.
	ECS / EKS	Managed Docker (ECS) and Kubernetes (EKS) container orchestration.
	Elastic Beanstalk	"Platform as a Service" that auto-provisions EC2 + load balancer + RDS for your app.
Storage & CDN	**S3** (Simple Storage Service)	Durable object storage for files, backups, static websites.
	EBS (Elastic Block Store)	Persistent block volumes for EC2.
	EFS (Elastic File System)	NFS file share that scales elastically across AZs.
	CloudFront	Global content-delivery network with edge caching & TLS.
Databases	**RDS** (Relational Database Service)	Managed MySQL, Postgres, MariaDB, SQL Server, Oracle.
	Aurora	AWS-optimised MySQL/Postgres compatible engine (RDS family).
	DynamoDB	Fully managed NoSQL key-value/document database with single-digit-ms latency.
	ElastiCache	Managed Redis and Memcached for low-latency caching.
Networking & Edge	**VPC** (Virtual Private Cloud)	Isolated network segment where you launch AWS resources.
	Route 53	Highly available DNS + traffic-routing service.
	API Gateway	Front door for REST/HTTP/WebSocket APIs with throttling, auth, caching.
	Global Accelerator	Anycast entry points that speed up TCP/UDP traffic to multi-region apps.
Security & Identity	**IAM** (Identity & Access Management)	Fine-grained permissions for users, roles, and services.

Category	Key Services	One-Sentence Purpose
	KMS (Key Management Service)	Create & manage encryption keys integrated with most AWS services.
	Secrets Manager / Parameter Store	Secure storage for API keys, DB credentials, config values.
Observability	**CloudWatch**	Metrics, logs, alarms, dashboards for AWS resources + custom sources.
	X-Ray	Distributed-tracing service for debugging microservices & Lambda flows.
Analytics & Big Data	**Athena**	Serverless SQL queries on data in S3.
	Glue	Serverless ETL and data catalog.
	Redshift	Petabyte-scale columnar data-warehouse.
Machine Learning	**SageMaker**	End-to-end ML platform—build, train, deploy models at scale.
	Rekognition / Transcribe / Comprehend	Pre-built computer-vision and NLP APIs.
Dev Tools & CICD	**CodeCommit**	Private Git repositories.
	CodeBuild, CodeDeploy, CodePipeline	Managed build servers, rolling deploys, and pipeline orchestration.
Migration & Hybrid	**DMS** (Database Migration Service)	Minimal-downtime migrations to AWS databases.
	Snowball / Snowcone	Physical devices to move terabytes of data into/out of AWS.

How to remember them quickly

1. **Compute trio:** EC2 (VMs), ECS/EKS (containers), Lambda (serverless).
2. **Storage trio:** S3 (objects), EBS (blocks), EFS (shared files).
3. **Database trio:** RDS/Aurora (SQL), DynamoDB (NoSQL), ElastiCache (in-memory).

Start with those nine, then branch into networking (VPC, Route 53) and security (IAM, KMS). Once they're familiar, exploring the rest feels far less overwhelming.

8 ▸ Quick Start Commands (Optional Hands-On)

```
# List available regions via CLI
aws ec2 describe-regions --output table

# Launch a t3.micro in chosen region (dry-run)
aws ec2 run-instances \
  --image-id ami-0c02fb55956c7d316 \
  --instance-type t3.micro \
  --region us-east-1 \
  --dry-run
```

Copy Explain

> `--dry-run` = *permission check only (no charge).*

9 ▸ Summary Table

Term	Elevator Pitch
Region / AZ	Independent data-centre clusters; design for multi-AZ HA
Edge Location	CDN nodes for low-latency content
Well-Architected	6-pillar checklist for healthy workloads
CAF	Roadmap for organisational cloud adoption
Shared Responsibility	Divide & conquer cloud security duties

Quiz Answers

1. **Edge Location**.
2. **Cost Optimisation** & **Performance Efficiency** (right-sizing affects cost).
3. **False** — Customers choose/maintain encryption settings and access controls.

AI TUTOR ACTIONS

Generate a practice quiz Summarize this unit Ask about this unit

Complete unit